

HW 3: Margin of Error and Matched Pairs

1) Your local gym wants to estimate the mean amount of time (in hours) that their members spend each week at their center. They would like a 95% confidence interval with a margin of error of 30 minutes. Based on previous data, they know that the population standard deviation is 2.2 hours. What is the smallest sample size that will allow this confidence interval to be built?

2) The number of hours college students spend studying per week follows a normal distribution. If you go three standard deviations out on either side of the mean, you get an interval of values from 10 hours to 34 hours. We want to estimate the mean study hours of a group of college students with 90% confidence and a margin of error of 2 hours. What is the sample size needed to get this confidence interval?

_____ 3) A 95% confidence interval for a population mean is determined to be 300 to 500. If the confidence is increased to 99% confidence while the sample statistics and sample size remain the same, which of the following are true?

I. The confidence interval for μ will become wider.

II. The critical value will increase.

III. The margin of error will decrease.

(A) I only (B) I and II only (C) II and III only (D) I and III only (E) I, II, and III

_____ 4) When constructing a one sample t interval for a population mean μ with a known standard deviation σ , which of the following changes will NOT change the width of the confidence interval?

I. Increasing the sample mean

II. Increasing the confidence level

III. Increasing the sample size

(A) I only (B) II only (C) I and II only (D) II and III only (E) I and III only

Use the following scenario for questions 5 – 7.

Research has shown that positive affirmations can have beneficial effects on academic performance and stress reduction. For example, studies have indicated that using self-affirmations can activate brain regions associated with positive emotion regulation and reward processing, which can lead to improved outcomes in stressful situations. A team of researchers set up a study to test this on 7 college students. They were given a “before” puzzle, and it was recorded how many minutes it took them to complete it. Then, they were given a two-hour break, where they listened to a playlist of positive affirmations. After the two hours, they took an “after” puzzle, which was similar to the first but changed a little. The number of minutes it took for them to complete the puzzle was recorded.

	Time to Complete the Puzzle						
Person	A	B	C	D	E	F	G
Before	54	57	68	43	42	49	66
After	52	50	67	44	30	38	61

_____ 5) Which of the following conditions must be met in order to use a t -procedure on these paired data?

- (A) The distribution of both before times and after times must be approximately Normal.
- (B) The distribution of before times and the distribution of differences (before – after) must be approximately Normal.
- (C) Only the distribution of before times must be approximately Normal.
- (D) Only the distribution of differences (before – after) must be approximately Normal.
- (E) All three distributions—before, after, and the difference—must be approximately Normal.

_____ 6) If μ_d represents the mean different in before – after, which of the following would be the appropriate hypothesis to test if positive affirmations make a student complete a puzzle faster?

- (A) $H_0: \mu_d > 0$ and $H_a: \mu_d = 0$
- (B) $H_0: \mu_d \neq 0$ and $H_a: \mu_d = 0$
- (C) $H_0: \mu_d = 0$ and $H_a: \mu_d < 0$
- (D) $H_0: \mu_d = 0$ and $H_a: \mu_d \neq 0$
- (E) $H_0: \mu_d = 0$ and $H_a: \mu_d > 0$

_____ 7) You run a one sample t test on the mean difference in before time – after time and obtain a p -value of 0.0155. If you significance level is 0.05, which of the following is correct?

- I. We are 95% confident that we can reject the null hypothesis.
- II. We have convincing evidence there is a positive difference in test scores.
- III. The probability of making a Type I error is 0.0155

- (A) I and III only
- (B) I and II only
- (C) II and III only
- (D) II only
- (E) I, II, and III